

GROUP WORK PROJECT # 2
GROUP NUMBER: 5178

MScFE 560: FINANCIAL MARKETS

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Student A - Open ended Mutual Fund: Vikas Gautam

Student B - ETFs: Salvatore Alessandro Chiruzzi

Student C - indexed annuity: Vikas Gautam & Salvatore Alessandro Chiruzzi (group of 2)

Step 1

Ques. 1:

1.a) Vikas Gautam: An open-ended, low-cost mutual fund that tracks a broad market index

Low running costs, minimal portfolio turnover, and wide market exposure are features of index mutual funds. These funds, irrespective of market conditions, track their benchmark index. such as the S&P 500 or NASDAQ 100 index. Since the first index mutual fund was established in 1976 by Jack Bogle, the founder of Vanguard, index investing has been the most popular type of passive investment. Open-ended low-cost mutual funds provides an easy and reasonably priced option for investors to get exposure to a diverse portfolio of stocks, bonds, and other securities

For retirement accounts, like individual retirement accounts (IRAs) and 401(k) accounts, index funds are thought to be the best core portfolio investments. The renowned investor Warren Buffett suggests index funds as a safe harbor for retirement savings. He said that instead of choosing particular businesses to invest in, the average investor would be better off buying all of the S&P 500 companies at the low cost of an index fund.

1. Performance:

It is observed that many index funds outperform actively managed funds in the long term, especially after fees and costs have been taken into account. Though, active investment funds perform better over shorter periods. for example, in the five years ending December, 2022, about 87% of large-cap funds in the U.S. returned less than the S&P 500 index.

Low-cost funds have the ability to significantly impact our long-term performance. We may be able to receive the same return from these at a lower cost as we would get from an actively managed funds with a similar theme because less of our investment is allocated to costs and more is invested.

The diversified combination of holdings assists in reducing the risks connected to individual equities and coordinate their performance with the direction of the market as a whole.

2. Fees:

The fund's expense percentage reflects the additional costs that investors must pay for fund management. Because index funds are handled passively, their expense ratios are typically lower. A team of analysts and active managers is not required, which reduces operating costs. Lower taxes for investors. Inexpensive index funds frequently have fees of just 0.05% or less, as opposed to the significantly higher fees of actively managed funds, which can reach 1.00%. For mutual funds, there are

apparently no transaction fees for holders. Based on cost comparisons, the majority of passive retail investors select index mutual funds over exchange-traded funds (ETFs).

Both commissions and trading fees costs are associated with index mutual funds. While some mutual funds may not have load fees, but could cost more than exchange-traded funds (ETFs) due to marketing and administrative costs.

3. Transparency:

As they track a market index, the holdings and details of an index fund are generally known and consistent. Transparency is maintained by low-cost, open-ended mutual funds that track indices of the entire market. They provide investors thorough and current information on their holdings, strategies to investing, and results. Because of this transparency, investors are able to choose their investments with information and knowledge.

4. Liquidity:

The pricing of mutual funds is done at the end of the day. So, index funds can only be traded at the set price point at the end of the trading day. Long-term performance of index mutual funds can be just as strong as that of ETFs, despite their lack of flexibility. An open-ended fund would have a single or multiple units of NAV at the end of the day at which all the transactions take place.

The accessibility of index mutual funds is another advantage that makes them perfect for many buy-and-hold investors. For instance, investors can buy index mutual funds directly from the fund or through their bank. An account at a brokerage is not required. Their success has been largely fueled by their accessibility.

Investors can simply enter and exit their positions because they are easily bought and sold on a stock exchange. Among the most actively traded index mutual funds worldwide, for instance, is the Vanguard Total Stock Market Index Fund (VTSAX), which sees more than 20 million of trading volumes per day on average.

index mutual funds being less liquid than ETFs and lack ETFs' intraday trading flexibility.

5. Professional Management:

Mutual funds are investment instruments that are pooled and overseen by professionals in money management. Low-cost, open-ended mutual funds that track a wide range of market indices gain from the knowledge of seasoned investing professionals. These experts keep a close eye on the pertinent market indices and periodically rebalance the funds' holdings to guarantee that they appropriately represent the makeup of the indexes.

Passive management, which leads to positive performance, usually proves to be true in the long term. Passively managed funds do not try to beat the market. Instead, their strategy aims to balance the overall risk and return of the market, assuming that the market always wins.

Index mutual funds are favored by many retail investors, or individual, non-professional investors. They enjoy their ease of use, shareholder services (including check writing and phone support), and investing alternatives that allow for automatic payments.

6. Investor Protections (e.g. Fiduciary responsibilities, SIPC, SEC, FINRA, etc.)

- Several investor safeguards apply to low-cost, open-ended mutual funds that track broad market indices, protecting the interests of investors. The Securities and Exchange Commission (SEC) and the Financial Industry Regulatory Authority (FINRA) provide oversight, fiduciary responsibilities, and SIPC insurance against broker-dealer failures.
- When advising or overseeing investments in open-ended mutual funds, financial professionals such as brokers and investment advisors have fiduciary duties. In addition to making sure that investment suggestions are appropriate based on clients' financial objectives, risk tolerance, and other pertinent variables, they must operate in the best interests of their customers.
- Since brokerage firms do not own open-ended mutual funds directly, SIPC does not offer coverage for them. Rather, the assets of investors in mutual funds are held by the custodian bank or the fund itself. However, in the event that the brokerage company holding the cash or securities fails, SIPC coverage might extend to those assets.
- The open-ended mutual funds business, is governed by the SEC. Mutual funds must register with the SEC and give investors comprehensive disclosures via prospectuses and periodic reports, among other papers. To safeguard the interests of investors, the SEC is in charge of managing fund registration, disclosure, operations, and compliance with securities laws.
- Mutual fund sales are subject to rules and regulations set by FINRA, which also oversees brokerage firms and brokers. FINRA regulates broker-dealers, but transactions involving mutual funds that are carried out through brokerage accounts are also subject to its rules. To safeguard investors, FINRA regulates advertising, suitability, sales tactics, and other facets of mutual fund transactions.

1.b) Alessandro: An Exchange-traded Fund that tracks the same broad market index.

Like stocks, exchange-traded funds (ETFs) are collections of assets that are exchanged on an exchange. ETFs are typically preferred by institutional passive investors. Since their introduction in the 1990s, exchange-traded funds (ETFs) have seen substantial growth. The real ownership of securities is not a component of ETFs. ETF prices may fluctuate in relation to the fund's net asset value (NAV) either at a premium or at a loss.

The novice investor can outperform the majority of financial pros by making investments in index funds.

1. Performance:

An exchange-traded fund can provide chances to deploy advanced trading tactics because it can be sold short and margined or leveraged. ETFs that track broad market indices may perform differently based on a number of variables, such as the underlying index's performance, fees and expense ratio, historical performance, dividend yield, tracking error, and market circumstances.

The total return of an ETF, which accounts for both price growth and reinvested dividends, is frequently used to gauge its performance. Total return, which takes into consideration both capital gains and dividend income, offers a thorough analysis of an ETF's performance over a certain time frame.

The general state of the market, which includes interest rates, geopolitical developments, economic considerations, and investor attitude, affects the performance of broad market index ETFs. Bull markets can see ETFs perform well as stock prices increase, whereas bear markets can cause ETF values to fall. Investors should therefore take into account the state of the market and how it affects the performance of ETFs.

2. Fees:

ETFs are generally less expensive and more tax-efficient than comparable mutual funds. Expenses like management fees and taxes are reduced with ETFs. Redeemable assets are not subject to redemption fees, unlike certain index mutual funds where investors are responsible for paying redemption costs each time they sell shares or units. Because ETFs are designed to have fewer taxable events than index funds, they are more tax efficient than the latter.

ETF transaction costs, like commissions, premiums and bid/ask spreads, should be carefully taken into account while creating a portfolio.

Comparable mutual funds often have higher expense ratios than exchange-traded funds (ETFs). This is partially due to the fact that they are exchange-traded, which means that brokers or the exchange carry the customary charges rather than in the case of mutual fund, which is responsible for the entire cost.

3. Transparency:

Most ETFs have a daily reporting requirement about their holdings. It makes it easier for the investor to stay informed about the precise investments being made with their money.

4. Liquidity:

Although index funds and exchange-traded funds (ETFs) are comparable in many aspects, ETFs are thought to be easier to buy and sell, like common equities on a stock exchange, they can be traded more easily than index funds and conventional mutual funds. By buying ETFs, they can skip the separate accounts and paperwork needed for mutual funds. ETFs are always available for purchase or sale during the trading hours.

ETFs offer an additional advantage in that a basic brokerage account can be used to invest in them due to their stock-like trading capabilities. They can be bought in modest quantities without the need for additional paperwork or rollover fees, and there is no requirement to have a separate account.

ETFs appeal to a wide spectrum of investors, including both active traders (like hedge funds) and passive investors (like institutional investors), since investors can enter or exit an ETF position whenever the market is open.

5. Professional Management

The key advantages institutional investors cite for using exchange-traded funds (ETFs) include superior liquidity management, transition management (from one manager to another), and tactical portfolio adjustments, all made possible by the ease, flexibility, and convenience of ETFs.

The only investors with the ability to issue or redeem new shares in an ETF are major institutional investors referred to as Authorized Participants (APs), who are large market makers. They transact with the ETF manager to create fresh shares of the ETF.

6. Investor Protections (e.g. Fiduciary responsibilities, SIPC, SEC, FINRA, etc.)

- The duty of financial professionals, such brokers or investment advisors, to behave in their clients' best interests is known as **fiduciary responsibility**. Financial professionals are required to follow fiduciary standards while suggesting or managing ETF investments. This involves making sure that the recommendations they make align with their customers' overall investment profile, risk tolerance, and financial objectives.
- Congress created SIPC, a nonprofit organization, to shield investors from losing money and securities held by brokerage companies in the case of a brokerage firm's bankruptcy or insolvency. Currently, SIPC offers limited coverage of up to \$500,000 per customer, per brokerage firm for securities (including up to \$250,000 for cash claims). It is crucial to remember that SIPC coverage does not offer protection against investment losses brought on by changes in the market or drops in the value of securities.
- The principal regulatory body in charge of the US securities market is the SEC. The SEC is mostly responsible for overseeing the disclosure, trading, and registration procedures of exchange-traded funds (ETFs). It is mandatory for ETF issuers to register their funds with the SEC and furnish comprehensive disclosures regarding the investment goals, tactics, hazards, costs, and outcomes of the fund. In order to guarantee adherence to securities regulations and safeguard the interests of investors, the SEC also carries out inspections and enforcement proceedings.
- The regulatory organization, FINRA is in charge of overseeing brokerage houses and brokers that conduct business in the US. For broker-dealers, FINRA sets guidelines and regulations pertaining to suitability, advertising, sales techniques, and supervision, among other things. In addition, FINRA offers services for investor education, arbitration, and mediation to settle conflicts that arise between investors and brokers or brokerage firms.

1.c) Vikas and Alessandro: An indexed annuity, which uses zero coupon bonds and options; the underlying of the options are the same broad market index.

Indexed annuities are a contract between the owner and insurance company. Their return is based on the performance of a certain equity index (e.g. S&P 500). Such products offer higher returns compared to fixed annuities and also provide the protection of the principal against market downturns. Typically, the return on this kind of product is capped (i.e. there is a maximum

return that an investor can earn). An indexed annuity using zero coupon bonds and option is a sophisticated product that in practice provides the investor the potential of a broad market index and at the same time a certain degree of capital protection.

1. Performance

The performance of an indexed annuity is connected with the performance of its underlying index. Usually there is a guaranteed minimum on the return, acting as a protection on the principal for investors in case of a negative market performance. This is balanced by the fact that the return is capped, meaning that there is a limit on the maximum profit an investor can make during highly positive market performance.

The use of zero-coupon bonds and options offer a good balance between risk and reward because options allow for a high return in case of good market performance, while zero coupon bonds ensure that the initial capital is returned at annuity maturity.

2. Fees

Fees are usually higher than other investment products such as ETFs and Mutual Funds. The fees' structure can be complex because it involves several layers of fees such as the cost of managing the annuity, fees on option purchase, and the implicit cost of the zero-coupon bonds. Given the complexity of this product, there could be also additional charges for administration and operational costs

3. Transparency

The complexity of this financial product requires very high transparency to possibly convince an investor. Calculating the return is complex because it involves crediting interest based on the performance of options and the amortization of the bonds. Additional transparency is required over the criteria for option selection, the indexing method. Also, the impact of volatility on return are important information that should be provided to investors

4. Liquidity

Due to the structured nature of this product, usually early withdrawal or liquidation is associated with high penalties and fees. Usually, these products have a lack of liquidity and this is to be carefully evaluated from investors requiring access to capital before maturity.

5. Professional Management

The expertise for these financial instruments is crucial especially in the selection and timing of options aligned with market prediction, the purchase of zero-coupon bonds that matches the annuity maturity profile and the rebalancing of these sub-instruments to optimize the performance.

6. Investor Protections (e.g. Fiduciary responsibilities, SIPC, SEC, FINRA, etc.)

For an indexed annuity the investor protection is complex because it reflects the hybrid nature of the product. Fiduciary responsibilities require the management to operate in the absolute interest of their investors. The SIPC do not cover insurance products like annuity. The securities components (i.e. options and zero-coupon bonds) may be protected with some mechanism, and the SEC ensure transparency over these components, while FINRA regulates the operation of brokers managing these securities. The insurance component of the annuity is typically regulated by national authorities

Ques 2. Let's examine open-ended mutual funds, exchange-traded funds, and indexed-annuities along 5 factors:

- Student A plays the role of a salesperson at the mutual fund: Vikas Gautam
- Student B plays the role of a financial advisor recommending the ETF: Alessandro
- Student C plays the role of an insurance salesperson recommending the Indexed Annuity: Alessandro and Vikas

2.7. Collateral-related risks: financing and credit

2.7.a Explain the credit risks, if any, of your product.

Student A - Open ended Mutual Fund:

These funds combine investor capital to purchase a range of securities, including bonds, stocks, or combinations of the two. At the close of each trading day, they issue and redeem shares at their net asset value, or NAV. If open-ended mutual funds purchase bonds or other fixed-income instruments from corporations or governments, so they run the risk of experiencing credit risk. The value of the mutual fund's holdings may drop if one of the securities' issuers defaults on payments, which would lower the fund's total value and the unit holders may suffer from credit loss. Also, the NAV is subject to price fluctuations which would be the factor for credit risk.

Student B - ETFs:

The fact that mutual funds and exchange-traded funds (ETFs) are legally independent of the management business is a crucial feature. They are set up as independent trusts, limited partnerships, or investment companies. This is significant because investors will still own the assets held by the fund, even in the event that the parent company of the exchange-traded fund (ETF) goes out of business.

ETFs and mutual funds are comparable in that they combine investor capital to purchase a diverse range of assets. But like stocks, exchange-traded funds (ETFs) are subject to daily price fluctuations as they are traded on an exchange. If ETFs use leverage (borrowing) to increase returns or own bonds or other fixed-income instruments with weaker credit ratings, they may be subject to credit risk. The value of the ETF could decrease if the issuer of these assets defaults or there is a fall in price.

Student C - indexed annuity:

Indexed annuity's structure is designed to minimize credit risk for the investor because the insurance company assumes most of this risk. The only credit risk in this financial instrument is related to the insurance ability to fulfill its obligations, and this is inherently unlikely.

2.7.b What institutions, if any, insure the payment of dividends/credit guarantees?

Student A - Open ended Mutual Fund:

Generally speaking, there are no dividend or credit guarantees associated with open-ended mutual funds. The Securities and Exchange Commission (SEC) in the US is in charge of regulating these funds, and the performance of the underlying securities owned by the fund determines the returns that investors receive. Investors in mutual funds assume the risk of possible losses in the event that the value of the underlying securities declines. Mutual funds do not guarantee returns.

Student B - ETFs:

ETFs often don't have dividend or credit guarantees. ETFs function differently from mutual funds and are subject to SEC regulation. Although certain exchange-traded funds (ETFs) may provide dividend yields in relation to the underlying securities they own, these yields are subject to change and are not assured. To meet their investment goals, ETFs may also employ derivatives or other financial instruments, which carries additional risk.

Student C - indexed annuity:

Insurance policies known as index-indexed annuities offer a mix of guaranteed minimum returns and rewards determined by the performance of a predetermined index, like the S&P 500. The financial stability of the issuing insurance firm is important even if indexed annuities are not usually directly exposed to credit risk like mutual funds and exchange-traded funds (ETFs) are. The annuity's capacity to meet its guaranteed minimum returns or other contractual commitments may be impacted if the insurance company goes bankrupt or defaults or shuts down.

The guarantees, including any pertaining to dividends or credited interest, are provided by the insurance company that issued the indexed annuity. State insurance departments oversee the regulation of indexed annuities in the US, and the insurance company issuing the policy must be creditworthy and financially sound. In the event of an insurance company's failure, state insurance guarantee associations may offer some protection, but the scope of coverage varies by state and is subject to specific limitations.

2.7.c What questions will A and B want to ask C about the credit guarantees for the annuity?

Student A - Open ended Mutual Fund:

Question for C: "Considering that performance can be variable, how do you make sure that the guarantees is robust for the annuity? In case of a long market downturn, are there any mechanism in place to protect the investor?"

Student B - ETFs:

Question for C: “Since the annuity is based on the insurer’s ability to protect the investor, how frequently does your company review the credit rating of the insurance? What happens if the rating of the insurance company is decreased? What do you do to make the portfolio rebalancing transparent”

2.8. Statistical related risks: volatility and correlation

Question 2.8.a

Student A - Open ended Mutual Fund:

Since all of the stocks in the index have medium to high correlation, they may still benefit from some degree of diversification. Diversification can lessen the volatility of any one stock's effect on the portfolio as a whole. Therefore, in comparison to buying individual equities, investing in a mutual fund might still offer some degree of risk protection.

Although stock correlation can help diversify portfolios, it does not always imply safety, and the relative safety of an open-ended mutual fund that follows such an index depends on a number of variables other than diversification such as market risk, concentration risk, tracking error because of factors like costs, trading fees, and choices made about portfolio management, investment objectives, risk tolerance, and time horizon. But if the index underperforms it will affect the mutual fund portfolio as a whole that follows the index raising factors for credit risk in terms of loss (of NAVs) or default.

Student B - ETFs:

If there is a medium to high correlation between the stocks in the index, it means that these stocks will probably move together when the market fluctuates. This reduces the diversification advantages that investors usually look for in broad market index exchange-traded funds.

The performance of the ETF is particularly susceptible to systematic risk when stocks in the index move in tandem as a result of correlated factors like market sentiment or economic conditions.

Due to the strong correlation between index components, the ETF's lack of diversification can exacerbate losses for investors during a market collapse or time of elevated volatility. There might not be much protection against significant market falls even if the ETF owns a wide range of stocks that react similarly.

Student C - indexed annuity:

When equities are medium to highly correlated indexed annuities are able to protect the principal for the investor. Typically, high correlation is a sign of market movements that affect an entire sector or geographical regions. Indexed annuities reduce this risk ensuring that also in a high volatility context the initial investment is protected.

Independently on the correlation between the underlying instruments, indexed annuities are a secure investment because the principal is decoupled from any market movement.

Question 2.8.b What questions does C want to ask A and B about down markets?

Student C - indexed annuity:

Question for A: What risk minimization strategies do you employ to protect the fund's performance from the impact of volatility? Does your fund adapt its strategy during significant market negative periods?

Question for B: ETFs are immediately impacted by market volatility. Do you consider your ETF to be resilient in times of high volatility compared to indexed annuities?

Question 2.8.c What questions do A and B want to ask C about the correlation between the participation rate and the equity market performance?

Student A - Open ended Mutual Fund:

Question for C: If the performance of the equity market changes significantly, how do you adjust the participation rate? What are the implications of this adjustment during periods of high volatility?

Student B - ETFs:

Question for C: What mechanism do you have in place to ensure that returns are competitive while avoiding exposing investors to excessive risk during negative market periods?

2.9. Magnifying risk factors: leverage and non-linearity

Question 2.9.a Explain how the indexed annuity has no downside (e.g. downside protection).

Student A - Open ended Mutual Fund:

Indexed annuities have a certain protection level against market declines compared to mutual funds, thanks to downside protection. This is a mechanism that protect the investor's principal also when the underlying index is performing below expectations. Moreover, unlike mutual funds, indexed annuity does not reflect the tracked market indices.

Student B - ETFs:

ETFs are immediately exposed to market, and in this sense indexed annuities are a good alternative. By offering downside protection it is ensured to the investor to not lose the principal in case of market downturns. Conversely indexed annuities also provide participation in market positive periods thanks to their underlying assets like options

Student C - indexed annuity:

The structure of indexed annuities protects investors from market downturns thanks to downside protection. Even if the broad market index tracks the market, the annuity at least

guarantees the return of the principal. This protection is also enabled by the combination of zero-coupon bonds maturing at their nominal value and the potential yield of options on the stock market. The investor is essentially experiencing growth possibilities without risk on the initial investment

Question 2.9.b Explain how the indexed annuity has an upside.

Student A - Open ended Mutual Fund:

Indexed annuities offer an upside thanks to a participation rate that enables investors to gain a percentage of the market growth. In mutual funds all of the market movements impact the investment, while in indexed annuities the growth is damped but at the same time the investor is shielded from downturns, making this an optimal alternative for more cautious investors.

Student B - ETFs:

The upside in indexed annuities is driven by the participation rate. This means that the benefit from market positive trends is capped to a certain limit. At the same time, it prevents from direct exposure to market losses.

Student C - indexed annuity:

In Indexed annuities the upside is made possible by leveraging the market growth through the participation rate. This means that the investor can enjoy market gains without exposing his principal

Question 2.9.c Explain the leverage of the participation rate.

Student A - Open ended Mutual Fund:

The participation rate specifies the share of the gain that the investor earns. For example, given a participation rate of 80% and an index gain of 10%, the investor is returned 8%.

Student B - ETFs:

The participation rate is a multiplier on returns based on the performance of the index. For example, considering a 60% participation rate, if the index grows by 10%, the index annuity gives 6% to the investor, and this directly links the gains to the performance of the underlying index.

Student C - indexed annuity:

The leveraging of the participation rate determines how much the investor benefits from gains of the index. It is a mechanism that allows the investor to earn a specified percentage of the index in case of positive performance.

Question 2.9.d From the previous question, review the correlation between the participation and equity market performance

Student A - Open ended Mutual Fund:

A high participation rate reinforces the correlation, and this allows investors to track with higher fidelity the market and benefit from potential upward movements. Conversely lower participation rate limits the possibility of the investor to gain from positive market movements

Student B - ETFs:

The participation rate determines what share of the index positive performance is recognized to the annuity. This establishes a direct (but moderated) link to market gains. If the market goes up also the potential return on the annuity increases, but only up to a certain percentage. In this way investors benefit from positive market movements while having a buffer against downturns.

Student C - indexed annuity:

The higher the participation rates the more the equity market performance impacts the gain of the investor. lower participation rates reduce the impact of market performance over the return of the annuity.

2.10. Frictional risk factors: liquidity

Mutual Funds

10.a.i The inflow of large capital might be forcing the fund managers to evaluate additional investment alternatives quickly, and this could potentially lead to non-optimal investment decisions. On the contrary, significant outflows could force managers to sell assets to a sub-optimal price, negatively affecting the fund's performance.

10.a.ii Mutual funds are traded at the end of the day based on their NAV (net asset value). For this reason, investor cannot make gains on price fluctuations during the day

10.a.iii Mutual funds disclose their holdings with a certain lag (quarterly typically). This delay can actually hide current investment decisions to the investor.

Exchange-traded fund (ETF)

10.b.i ETFs can be traded intraday, and this can lead to price deviation from the underlying asset. This discrepancy can actually affect the performance of the ETF, leading to premiums or discounts on the ETFs relatively to the NAV

10.b.ii ETFs are associated with lower management fees and brokerage commissions compared to a mutual fund.

10.b.iii Sometimes, when ETFs get dividends from the stocks they own, they hold the cash for a while before sharing it with the investors. This cash, doesn't earn extra gain during this holding time.

Indexed Annuity

10.c.i Indexed annuities include substantial early withdrawal penalties, and this usually lock investors for long periods (5 to 10 years for example). This has of course a big impact on the liquidity of these assets

10.c.ii Investors have no control on the participation rate set by the insurance company. Changes on this rate can of course affect the potential returns.

10.c.iii Fee structure for indexed annuity can be extremely complex. These fees (administrative, expenses, mortality, etc.) are not always clearly disclosed making challenging for the investors to understand the implications of their annuity

Step 3 - Group Report

Introduction: This document provides a comprehensive comparison of 3 investment cars: open-ended mutual finances, exchange-traded budget (ETFs), and indexed annuities, specializing in their responses to collateral-associated dangers, statistical-related dangers, magnifying threat elements, and frictional elements. Each investment product gives specific capabilities and advantages, tailored to satisfy unique investor wishes and threat tolerances. Our evaluation synthesizes perspectives from 3 roles: a mutual fund salesclerk (Student A), an ETF economic guide (Student B), and a coverage salesclerk recommending indexed annuities (Student C).

Collateral-Related Risks: Credit Risk Mutual Funds and ETFs exhibit low credit chance because of their varied and portfolio nature, minimizing the impact of any single safeties or stocks failure. While nor is insured towards market losses, both are regulated through the SEC, making sure investor protection via transparency and disclosure. Indexed Annuities offer a different technique to credit risk, focusing on the insurer's potential to satisfy it duties. The predominant investment is covered, with nation guarantee institutions presenting a safety net, albeit inside limits. Mutual fund and ETF representatives are keen on knowledge of the mechanisms listed annuities employ to manage credit score risks, in particular the function of insurance guarantee institutions.

Statistical-Related Risks: Correlation Mutual Funds and ETFs tracking a huge market index are challenge to market-huge dangers, with their performance closely mirroring the index. High correlation among equities can divulge investors to systemic dangers at some stage in downturns. Indexed Annuities provide a layer of protection towards marketplace volatility through guaranteed minimum returns, making them much less prone to losses for the duration of intervals of excessive market correlation. The discussion highlights curiosity about how each product mitigates losses in the course of market corrections, with a specific interest in the annuity's participation fee's impact on returns in unstable markets.

Magnifying Risk: Indexed Annuities differ from other investments in the relationship between the participation rate and the market conditions. The participation rate aligns the profit potential of the investor with the appreciation instead of increasing risk of participation in the market with the market downturn. In this regard, the participation rate leverage effect provides a certain degree of nonlinearity to returns, which is a risk reward balance that has a number of associated services.

Leverage is just about applicable to the latter types of fund, because mutual and ETFs do aim at the market's tracking, rather than at outperforming it. Still some specialized leveraged ETFs apply the derivative financial instrument so as to multiply return, which in turn multiplies the risk of loss.

Investors tend to be more flexible when it comes to participating in the market returns in market-linked annuities, taking into consideration the nonlinear nature of returns and the correlation between the rate of participation and market performance, helping to understand how investing in such instruments can be a way of balancing risk and reward.

Frictional Factors: Besides facing unidirectional flows of large capital volume problem, mutual funds lack intraday trading mechanism which is inhibiting their possibility to respond to market situations promptly. Index funds managers are considered by regulators as hedge funds and investors are mostly retail investors. Periodic disclosure of holdings may also affect the transparency. The evaluation of price and trading is done at the end or closing of the trading day. Dividends can be reinvested as growth mode or paid back to investors.

ETFs possess greater liquidity than stocks since they are traded over-the-counter but are also the product of the fast-flowing market influencing the price and the cost of trading and fees. Dividend distribution can be declared to be another level of inconsistency.

The fee to withdraw the indexed annuities entails charge and limited allocation rates, as well as unclear fee plans, affecting the flexibility and the openness of the market that easily becomes unclear.

Conclusion The almost similarity each instrument in these groups of mutual funds, ETFs and indexed annuities has in term has different advantages and drawbacks. Mutual funds and Exchanged Traded Funds (ETFs) offer the markets spread coupled with different liquidity and investor information levels. Indexed annuities on the other hand, guarantee principal protection and give structured returns. While it is important to be familiar with the details of asset classes such as credit risks, market correlation, leverage effects and rules, this is essential for anyone who likes to invest and choose from a variety of options going forward in their future investments. This analysis clearly shows the need for consistent and purposeful allocation of the capital to adjust it according to personal priorities, risk profile, and monetary accessories.

2.11 Regulation

2.11.1 Mutual fund

- The document explores the US mutual fund market, emphasizing the sector's explosive expansion and the rise in mutual fund companies, which aided in the scandal by giving criminals access to a bigger pool of assets to steal from. It also covers the effects of deregulation and the rise of cheap brokerage firms in the US, which lowered commissions and enhanced competition in the securities industry.
- The US mutual fund controversy is discussed in the document, with special attention paid to how it started in 2003 and how the Attorney General's office in New York handled the ensuing inquiry. It describes how a whistleblower named Noreen Harrington informed authorities

about illicit trading going on between mutual fund companies and institutional clients, which prompted a larger probe into the mutual fund business. It tells about late trading, use of omnibus accounts, lack of transparency, conflict of interests, illegal trades and market timing for e.g. "Betting on a horse race after the horses have crossed the finish line" is how Spitzer put these activities.

- Drawing comparisons to the mutual fund crisis in the United States, the paper looks at the structural and cultural elements that might lead to financial fraud in Japan. It draws attention to the effects of Big Bang deregulation in Japan, the absence of openness, and the impact of Japanese corporate culture on regulatory reactions and whistleblowing. The study highlights the possible ramifications for the financial system in Japan and highlights the necessity of regulatory oversight and cultural sensitivity in order to stop financial exploitation.
- Several regulations pertaining to the mutual fund business are mentioned. These include the 1940 Securities Act, which stipulates that only 40% of the board's directors must be independent from the fund-management company, the elimination of fixed commissions from the London securities market, and the need for an independent chairperson and board of directors for mutual fund companies. The SEC has also made an effort to combat the lack of transparency by enacting regulations like rule 22c-2, which eliminates brokerage companies ability to use omnibus accounts for mutual fund investments and gives mutual funds access to additional information about investor transactions. The SEC has also given thought to requiring that seventy-five percent of the board's directors be independent. Suggesting that a key factor in determining whether and how funds react to the arbitrage problem is fund governance. The goals of these legislation are to better oversee the mutual fund business, address conflicts of interest, and increase transparency. Brokerage businesses were also forced to disclose the identities of their clients. Additionally, the brokerage must consent to abide by the mutual fund's guidelines prohibiting or limiting any shareholder's ability to make further purchases or exchanges should that shareholder transgress the market timing guidelines of the fund.
- Order tickets are susceptible to manipulation, as demonstrated by the mutual fund fraud, as is a time-stamping system. Orders might be manually manipulated by brokers with access to the time stamping machine, allowing them to backdate orders. Additionally, colluding brokers have the ability to reject orders that do not benefit the investor. A paper copy of the order ticket as a backup and an automated electronic system that delivered orders to clearinghouses for processing would address a number of issues and increase the effectiveness of the current system.
- In the late 1990s, Japanese Prime Minister Hashimoto introduced the "Big Bang" initiative, which sought to reform and de-regulate six areas of the Japanese economy. The idea of mass de-regulation was inspired by astrophysics, as the "Big Bang" was a large, singular event that recreated the Japanese economic universe. The Big Bang initiative also affected the Japanese version of the mutual fund market, investment trusts, by bringing transparency, enhance the system of regulation and liberalizing laws regarding who could sell these investments. Banks

have been granted the ability to sell their own investment trusts, making them more accessible to the public and potentially increasing their popularity as investment vehicles.

- The lack of information in trusts gave mutual funds the ability to abuse investors without their knowledge. Two factors have contributed to the increase in mutual fund abuse in Japan: the direct influence of commission liberalization, which has resulted in reduced commissions and better trade conduct, and the indirect effect of increased competition from discount brokerage firms. The deregulation of the investment trust industry may not always be effective in preventing fraud, as it may not fully address the issues faced by Japanese investors. Japanese investment strategy may limit potential investment trust abuse, as Japanese investors established relationships and provide monetary insurance through their investments. The slow deregulation of the Japanese system also helped in a way, as only five firms were approved to expand their business operations during the first year under the deregulated system.
- Financial fraud is more significant when endemic to a system, as seen in the American mutual fund scandal. Japan's competitive environment and less transparency may magnify the issue, as the Japanese culture's adherence to social and business hierarchies decreases the likelihood of early fraud detection. This creates a crime-facilitative environment, potentially leading to more damage over time. However, a lack of access to large pool of funds may discourage abusive investors.

2.11.2 Exchange Traded Funds (ETF)

- Synthetic ETFs can outperform physical ETFs, as investors' needs vary by region and tax jurisdiction. Synthetic ETFs use derivatives or swaps to replicate an index, like the iShares S&P 500 Swap UCITS. Currently, only 1,020 ETFs are synthetic, accounting for 11% of the ETF universe. 67% are in EMEA, 23% in the Americas, and 10% in the Asia-Pacific. Regulations governing the use of derivatives are frequently enacted. Examples of such regulations include counterparty risk management for synthetic ETFs and collateral requirements.
- The debate over synthetic vs. physical ETFs has pros and cons, with synthetic being preferred for replication, while some ETFs may have tax advantages, others as in oil ETFs comes with high storage and transport costs.
- Synthetic S&P 500 swaps versions, like iShares' I500 LN, outperformed physical S&P500 ETFs, one-year performance by 47 basis points.
- Swap-based ETFs offer lower trading differences due to lower withholding-tax drag. EMEA leads with 70% synthetic ETFs, partly due to UCITS regime tax benefits, while America has 30%, mainly in commodities.
- Performance of S&P 500 synthetic ETFs varies based on swap counterparty, gross or net index, management cost, and fee structure. Net benchmark swaps may be more volatile and cause higher tracking error. Despite Invesco being the cheapest, iShares outperforms all five ETFs.
- ETFs are regulated investment products, and the competent financial authorities in the areas in which they operate have regulatory authority over both synthetic and physical ETFs. The

Financial Conduct Authority (FCA) in the UK, the Securities and Exchange Commission (SEC) in the US, and other comparable organizations may fall under this category.

- Since physical exchange-traded funds (ETFs) mirror the underlying index directly, they typically provide more plain transparency. In order to make sure investors are aware of the dangers, authorities may impose additional disclosure requirements on synthetic exchange-traded funds (ETFs), which may have more complex structures.
- Market manipulation, insider trading, and preserving orderly markets are a few of the topics that regulations may touch on in relation to ETF trading. To guarantee honest and open trading, these rules apply to both physical and synthetic ETFs.
- By making sure that ETFs are managed in the best interests of investors, give accurate information, and abide by all applicable laws and regulations, regulatory authorities hope to protect investors. This covers the disclosure requirements for prospectuses, reporting guidelines, and continuous fund management monitoring.

2.11.3 Indexed Annuity

The document from St. John's University School of Law Scholarship Repository, titled "Suitability Obligations Applicable to Securities and Annuities," authored by Christine Lazaro and Benjamin P. Edwards, discusses the complex landscape of regulatory obligations brokers face when recommending securities and annuities to customers. The article addresses the intricacies of these obligations, emphasizing the variability depending on the product recommended, and sheds light on the overlapping and multifaceted regulatory framework brokers navigate, involving both federal and state oversight.

Crisis and Challenges

The crisis highlighted in the document revolves around the suitability obligations of brokers when making recommendations to customers, specifically in the context of securities and annuities. These obligations become particularly complex when brokers deal with products like variable and equity-indexed annuities, which straddle the line between securities and insurance products. The document outlines the confusion and potential for misinterpretation of regulations, which can lead to inadequate protection for investors and challenges in enforcing compliance among brokers.

Proposed Regulations and Guidelines

The document delves into the detailed regulatory framework designed to address these challenges, focusing on the following key areas:

- **FINRA Rule 2111 (Suitability Rule):** It mandates brokers to have a reasonable basis to believe that a recommendation is suitable for a customer based on a comprehensive understanding of the customer's investment profile. The rule emphasizes a broker's duty for fair dealing and ethical sales practices, ensuring recommendations align with the customer's best interests.
- **Regulatory Notices on Suitability:** FINRA has issued several notices to clarify aspects of Rule 2111, such as the definition of "recommendation," the scope of "investment strategy," and the

identification of who qualifies as a customer. These notices aim to provide brokers with clearer guidance on their obligations and the expectations for compliance.

- **Broker's Obligations When Recommending Annuities:** The article distinguishes between different types of annuities (fixed, variable, and equity-indexed) and their respective regulatory frameworks. It highlights additional obligations under FINRA Rule 2330 for variable annuities, including the need to inform customers about the annuities' features and potential risks.
- **Equity-Indexed Annuities:** The document explores the regulatory ambiguity surrounding equity-indexed annuities, which have characteristics of both insurance and securities. It discusses the historical and legal context that has led to ongoing debates about whether these products should be regulated as securities or insurance.

NAIC Model Suitability Regulation: In response to the evolving landscape of annuity products and to enhance consumer protection, the National Association of Insurance Commissioners (NAIC) developed the Suitability in Annuity Transactions Model Regulation. This regulation sets forth suitability standards for annuity recommendations, requiring brokers to ensure that their recommendations fit the consumer's financial situation, needs, and objectives.

The crisis outlined in the document stems from the complexity of financial products and the dual nature of some annuities, leading to a regulatory environment that can be difficult for brokers to navigate. The proposed regulations and guidelines aim to clarify these obligations, ensuring that brokers' recommendations align with the best interests of their customers, thus fostering a fairer and more transparent marketplace.

Enhanced Investor Protection

The crisis at the heart of the document reflects a broader concern within the financial industry: ensuring that investors, particularly retail consumers, are adequately protected from unsuitable investment advice. The complexities introduced by innovative financial products, like variable and equity-indexed annuities, exacerbate the challenge. These products, which blend characteristics of both securities and insurance, necessitate a nuanced understanding from brokers to ensure recommendations are truly in the best interests of their clients.

Regulatory Evolution and Response

The evolution of financial products has prompted regulatory bodies like FINRA and the NAIC to adapt and refine their approaches to investor protection. The introduction of FINRA Rule 2111 was a significant step towards formalizing the expectation that brokers must not only understand the products they recommend but also the unique financial situation and goals of their customers. This rule, along with subsequent clarifications and additional regulations like FINRA Rule 2330 for variable annuities, represents a concerted effort to elevate the standard of care in the financial advisory sector.

The Role of State Regulation and the NAIC Model Regulation

The article underscores the significant role state regulation plays in governing the sale and recommendation of annuity products. The variability in state adoption of the NAIC Model Suitability Regulation points to a fragmented regulatory landscape that can potentially leave gaps in investor protection. The Harkin Amendment, as highlighted in the document, attempts to bridge these gaps by

setting federal standards for the treatment of certain annuity contracts, including equity-indexed annuities, under the Securities Act, provided they meet specific criteria.

Challenges and Opportunities

Despite these regulatory advancements, the document alludes to ongoing challenges in ensuring that these rules are effectively enforced and that they keep pace with the innovation in financial products. Brokers and financial advisors face the dual challenge of navigating a complex regulatory environment while striving to meet the evolving needs and preferences of their clients. This dynamic landscape presents both challenges and opportunities for the industry to improve transparency, trust, and ultimately, the investor experience.

Future Directions

Looking forward, the financial industry and regulatory bodies must continue to work collaboratively to refine and adapt regulations that not only address the current complexities but are also flexible enough to accommodate future innovations. This includes ongoing education for brokers on the nuances of financial products and regulatory requirements, as well as enhanced oversight mechanisms to ensure compliance. Moreover, there's a growing need for leveraging technology to aid in the compliance process, making it more efficient for brokers to align their recommendations with the suitability standards set forth by regulatory authorities.

Thanks.